

## INDIANAPOLIS EAGLE CREEK, IN, USA

WMO: 724384

Lat: 39.825N

Lon: 86.296W

Elev: 823

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53842

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |            |            | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>2.6 | (c)<br>8.2 | (d)<br>-6.0                   | (e)<br>4.1 | (f)<br>4.7 | (g)<br>-1.3 | (h)<br>5.3 | (i)<br>9.5 | (j)<br>24.2           | (k)<br>26.6 | (l)<br>21.4 | (m)<br>27.5 | (n)<br>6.9            | (o)<br>300 | (p)<br>0.454 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>16.9            | (c)<br>90.6     | (d)<br>74.3 | (e)<br>88.6 | (f)<br>73.3 | (g)<br>86.5 | (h)<br>72.2 | (i)<br>77.4         | (j)<br>86.9 | (k)<br>75.9 | (l)<br>84.7 | (m)<br>74.7 | (n)<br>83.0 | (o)<br>7.4           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.4                            | 132.4 | 82.8 | 73.0 | 126.1 | 81.4 | 72.0 | 122.1 | 80.4 | 41.2          | 86.9 | 39.9 | 85.0 | 38.6 | 83.0 | 82.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |          | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|----------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |          | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max      | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 19.9          | (b) 17.6 | (c) 15.8 | (e) -3.6                   | (f) 93.5 | (g) 6.5            | (h) 3.0  | (i) -8.3                                           | (j) 95.6  | (k) -12.1  | (l) 97.3  | (m) -15.8  | (n) 99.0  | (o) -20.5  | (p) 101.2 |    |    |
| (4) 19.9          | (5) 17.6 | (6) 15.8 | (7) -4.3                   | (8) 79.3 | (9) 6.4            | (10) 1.8 | (11) -8.9                                          | (12) 80.6 | (13) -12.7 | (14) 81.7 | (15) -16.3 | (16) 82.7 | (17) -20.9 | (18) 84.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 53.7   | 28.3  | 31.6  | 42.6  | 54.2  | 64.0  | 72.6  | 75.6  | 74.3  | 67.9  | 55.5  | 43.5  | 33.0  |
|                                                                               | DBStd       | 18.97  | 11.94 | 11.49 | 11.11 | 9.28  | 8.13  | 5.82  | 5.33  | 4.85  | 7.10  | 8.93  | 9.51  | 10.62 |
|                                                                               | HDD50       | 2368   | 677   | 524   | 286   | 61    | 4     | 0     | 0     | 0     | 44    | 238   | 534   |       |
|                                                                               | HDD65       | 5247   | 1137  | 935   | 696   | 340   | 120   | 12    | 1     | 2     | 51    | 317   | 644   | 992   |
|                                                                               | CDD50       | 3726   | 5     | 9     | 58    | 187   | 439   | 677   | 793   | 754   | 537   | 215   | 45    | 7     |
|                                                                               | CDD65       | 1129   | 0     | 0     | 3     | 16    | 90    | 239   | 329   | 291   | 138   | 23    | 0     | 0     |
|                                                                               | CDH74       | 9531   | 0     | 0     | 20    | 137   | 739   | 1958  | 2999  | 2364  | 1140  | 172   | 2     | 0     |
|                                                                               | CDH80       | 3035   | 0     | 0     | 1     | 13    | 172   | 629   | 1077  | 746   | 359   | 38    | 0     | 0     |
| Wind                                                                          | WSAvg       | 6.8    | 8.2   | 8.1   | 8.1   | 8.3   | 6.6   | 5.8   | 5.3   | 5.0   | 5.2   | 6.3   | 7.2   | 7.6   |
| Precipitation                                                                 | PrecAvg     | 44.1   | 3.0   | 2.6   | 3.5   | 4.5   | 4.7   | 5.2   | 4.3   | 3.2   | 3.4   | 3.1   | 3.4   | 3.3   |
|                                                                               | PrecMax     | 55.2   | 9.6   | 5.9   | 6.4   | 8.3   | 9.5   | 10.8  | 11.3  | 6.7   | 9.1   | 7.9   | 8.0   | 8.0   |
|                                                                               | PrecMin     | 33.4   | 0.9   | 1.0   | 0.9   | 1.8   | 1.8   | 0.0   | 0.9   | 0.4   | 0.4   | 0.9   | 0.6   | 1.1   |
|                                                                               | PrecStd     | 6.6    | 1.8   | 1.3   | 1.5   | 1.7   | 1.9   | 2.2   | 2.7   | 1.5   | 2.2   | 1.7   | 1.7   | 1.6   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 62.8  | 66.5  | 77.0  | 81.9  | 88.1  | 91.3  | 95.4  | 92.5  | 91.3  | 84.8  | 72.1  | 63.3  |
|                                                                               |             | MCWB   | 57.6  | 56.2  | 62.0  | 64.1  | 70.4  | 73.7  | 76.7  | 74.6  | 71.6  | 69.4  | 59.3  | 56.7  |
|                                                                               | 2%          | DB     | 55.6  | 60.5  | 70.8  | 78.6  | 84.2  | 88.8  | 90.9  | 89.7  | 88.1  | 79.0  | 66.8  | 57.3  |
|                                                                               |             | MCWB   | 51.2  | 54.2  | 58.8  | 62.2  | 68.2  | 72.5  | 75.9  | 74.1  | 71.6  | 66.0  | 57.6  | 53.6  |
|                                                                               | 5%          | DB     | 51.1  | 54.3  | 65.0  | 74.3  | 81.6  | 86.5  | 88.5  | 86.9  | 84.1  | 74.2  | 63.1  | 53.1  |
|                                                                               |             | MCWB   | 47.0  | 48.9  | 55.4  | 60.2  | 67.0  | 71.6  | 74.6  | 72.9  | 69.8  | 63.5  | 55.8  | 49.4  |
|                                                                               | 10%         | DB     | 45.4  | 48.7  | 60.8  | 70.3  | 78.6  | 83.8  | 86.1  | 84.1  | 80.8  | 70.7  | 58.6  | 47.9  |
|                                                                               |             | MCWB   | 41.4  | 43.0  | 53.0  | 58.8  | 66.4  | 70.0  | 73.2  | 71.5  | 67.8  | 61.8  | 51.8  | 44.4  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 58.3  | 59.4  | 63.6  | 67.9  | 73.2  | 77.0  | 79.8  | 78.3  | 75.7  | 72.0  | 63.3  | 58.6  |
|                                                                               |             | MCDB   | 62.5  | 64.4  | 74.1  | 78.3  | 82.5  | 87.5  | 89.7  | 88.4  | 85.3  | 80.3  | 69.6  | 62.1  |
|                                                                               | 2%          | WB     | 52.9  | 54.6  | 60.4  | 65.1  | 71.1  | 75.2  | 78.1  | 76.5  | 73.8  | 69.1  | 59.8  | 54.1  |
|                                                                               |             | MCDB   | 54.9  | 58.3  | 68.5  | 73.3  | 80.9  | 85.0  | 87.5  | 85.6  | 83.1  | 75.9  | 64.6  | 56.9  |
|                                                                               | 5%          | WB     | 47.7  | 49.7  | 57.7  | 62.6  | 69.4  | 73.7  | 76.8  | 75.2  | 72.1  | 66.2  | 56.6  | 49.5  |
|                                                                               |             | MCDB   | 50.4  | 55.3  | 63.6  | 70.5  | 78.2  | 83.3  | 85.7  | 83.2  | 80.5  | 72.2  | 62.0  | 52.6  |
|                                                                               | 10%         | WB     | 41.5  | 43.4  | 53.7  | 60.2  | 67.7  | 72.2  | 75.2  | 73.8  | 70.4  | 63.0  | 52.6  | 44.9  |
|                                                                               |             | MCDB   | 45.8  | 48.6  | 60.0  | 68.2  | 75.8  | 80.8  | 83.3  | 81.2  | 77.6  | 68.6  | 58.0  | 47.4  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.6  | 15.5  | 17.6  | 19.3  | 18.3  | 17.4  | 16.9  | 17.3  | 19.1  | 18.2  | 16.2  | 13.3  |
|                                                                               |             | MCDBR  | 19.6  | 22.1  | 23.4  | 23.7  | 22.0  | 20.0  | 18.6  | 19.1  | 22.7  | 22.7  | 21.7  | 18.2  |
|                                                                               |             | MCWBR  | 16.7  | 17.8  | 14.6  | 12.9  | 10.1  | 9.0   | 7.4   | 7.9   | 9.0   | 11.8  | 14.9  | 15.7  |
|                                                                               | 5% WB       | MCDBR  | 18.4  | 20.6  | 20.5  | 19.3  | 18.1  | 17.4  | 16.6  | 16.3  | 18.2  | 18.1  | 19.0  | 17.7  |
|                                                                               |             | MCWBR  | 17.1  | 18.2  | 13.3  | 12.6  | 9.5   | 8.8   | 7.8   | 8.1   | 8.4   | 10.4  | 15.4  | 16.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.302  | 0.324 | 0.347 | 0.404 | 0.438 | 0.440 | 0.455 | 0.452 | 0.417 | 0.359 | 0.320 | 0.300 |       |
|                                                                               | taud        | 2.433  | 2.351 | 2.369 | 2.219 | 2.174 | 2.208 | 2.168 | 2.163 | 2.245 | 2.407 | 2.465 | 2.456 |       |
|                                                                               | Ebn at Noon | 275    | 283   | 288   | 276   | 268   | 267   | 261   | 259   | 260   | 267   | 266   | 266   |       |
|                                                                               | Edh at Noon | 26     | 32    | 35    | 43    | 47    | 45    | 47    | 46    | 39    | 30    | 25    | 23    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 556    | 818   | 1155  | 1503  | 1716  | 1955  | 1910  | 1739  | 1433  | 1002  | 667   | 466   |       |
|                                                                               | RadStd      | 59     | 91    | 99    | 124   | 156   | 119   | 112   | 101   | 116   | 96    | 73    | 52    |       |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +193  | N/A   |

Nomenclature: See separate page